

EXPERIMENT – 7

DESIGN AND SIMULATION OF FET AMPLIFIER USING LT SPICE SIMULATOR

OBJECTIVES

The experiment is divided into three engineering laboratory model tests:

1. Verification of Voltage Gain
2. Determination of Lower and Upper Cutoff Frequencies
3. Frequency Response and Bandwidth Analysis

EXPERIMENT – 7.1

ANALYSIS OF VOLTAGE GAIN FOR FET AMPLIFIER

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. To design a Common Source FET amplifier
2. To simulate the amplifier circuit using LTspice.
3. Measure voltage gain experimentally.
4. Compare theoretical and practical gain values.
5. Understand transistor amplification principles.
6. Analyze the effect of transistor parameters on gain.

Aim

To design and verify the voltage gain of a FET.

Theoretical Background:

A Field Effect Transistor (FET) is a **voltage-controlled semiconductor device** having high input impedance and low power consumption. In a **Common Source Amplifier**:

- Input is applied to the gate.
- Output is taken from the drain.
- Source acts as the common terminal.
- Output is amplified and is **180° out of phase** with the input.
- The source resistor provides bias stabilization.
- The bypass capacitor increases AC gain.

The voltage gain is approximately

DESIGN FORMULAE

$$A_v = \frac{R_D}{r_s}$$

or

$$A_v = g_m(R_D || R_L)$$

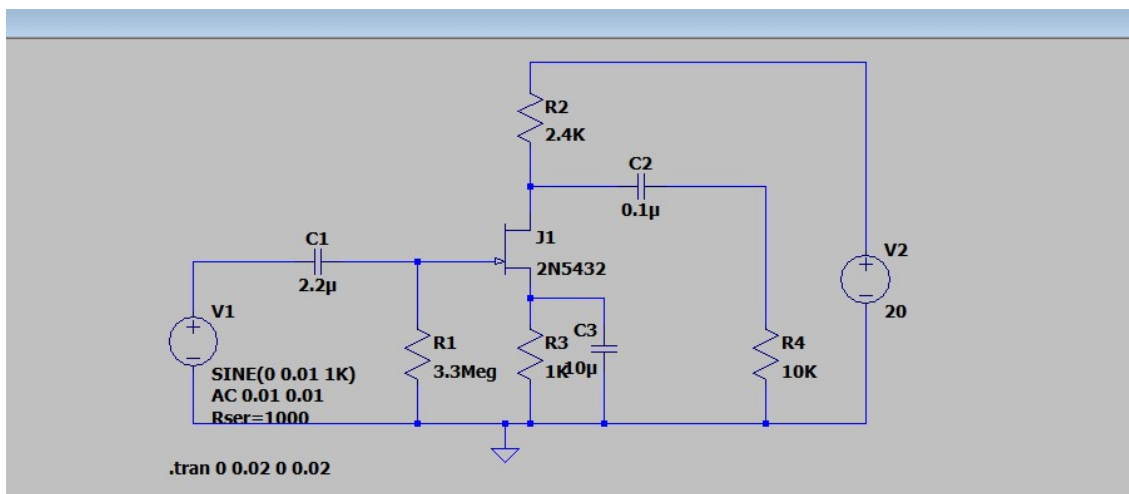
where

- g_m = Transconductance
- R_D = Drain resistance
- R_L = Load resistance

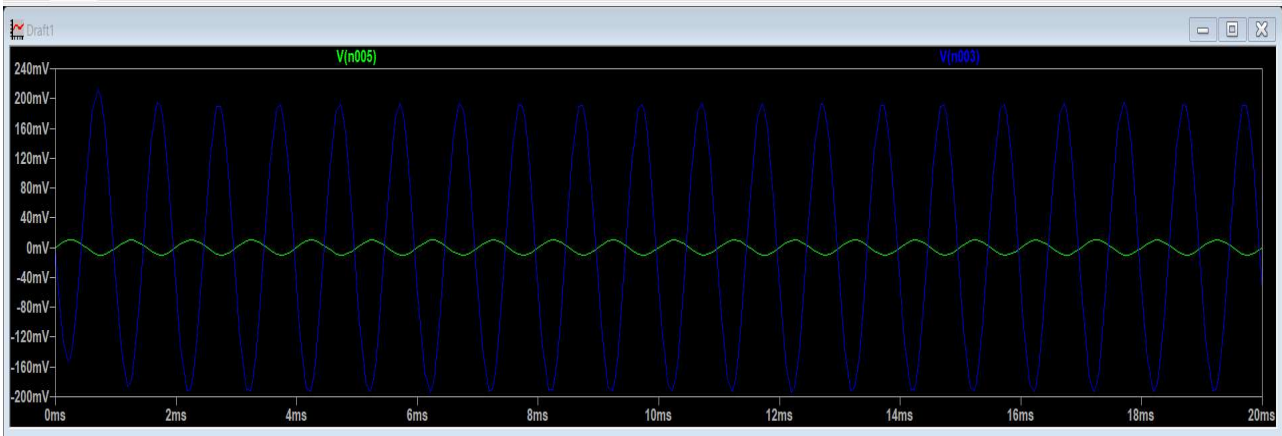
Design Procedure

1. Select the FET device (e.g., 2N5457 or 2N7000).
2. Choose the supply voltage (typically 12 V).
3. Select R_D and R_S to obtain a suitable operating point.
4. Choose R_G to provide proper gate bias.
5. Connect coupling and bypass capacitors.
6. Draw the circuit in LTspice.
7. Verify all component values.

Circuit diagram:



Model Graph:



Tabulation:

S.no	INPUT VOLTAGE Vi (mV)	OUTPUT VOLTAGE V0 (mV)	GAIN (dB)
1.	10.43	204.47	25.8469
2.	9.07	190.81	26.4599
3.	11.80	190.81	24.1744
4.	11.80	190.81	24.1744
5.	11.43	183.98	16.0962

RESULT

The midband voltage gain of the common emitter amplifier was verified successfully.

EXPERIMENT – 7.2

DETERMINATION OF LOWER AND UPPER CUTOFF FREQUENCIES FOR FET AMPLIFIER

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Identify cutoff frequencies.
2. Calculate bandwidth.
3. Understand frequency limitations of amplifiers.
4. Interpret gain-frequency characteristics.

Aim

To determine the lower and upper cutoff frequencies of a CE amplifier.

THEORETICAL BACKGROUND

At low frequencies:

Coupling capacitor reactance increases.

Emitter bypass capacitor becomes ineffective.

Therefore gain decreases.

At high frequencies:

Internal transistor capacitances dominate.

Gain decreases due to capacitive effects.

Cutoff frequencies are defined at the points where gain becomes:

$$0.707 A_{v(max)}$$

or

$$-3dB$$

from the maximum gain.

DESIGN FORMULAE

Lower Cutoff Frequency

$$f_L$$

Upper Cutoff Frequency

$$f_H$$

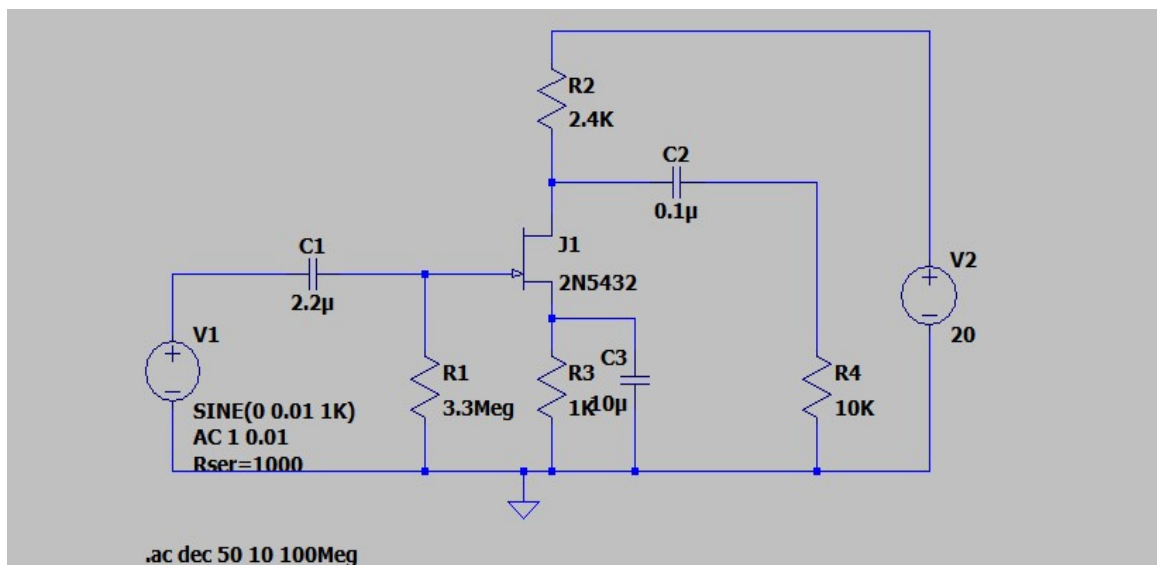
Bandwidth

$$BW = f_H - f_L$$

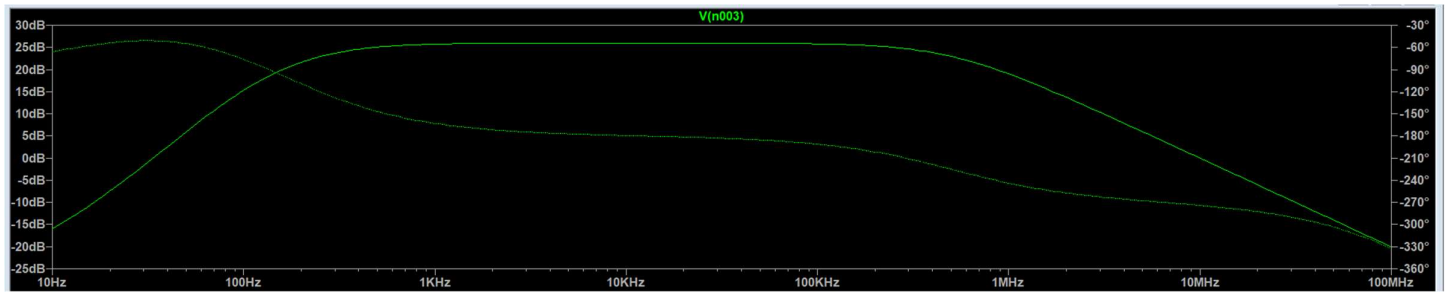
EXPERIMENTAL PROCEDURE

1. Set input voltage constant.
2. Start at 20 Hz.
3. Measure output voltage.
4. Increase frequency gradually.
5. Determine gain at each frequency.
6. Find frequencies corresponding to -3 dB points.
7. Calculate bandwidth.

Circuit diagram:



Model Graph:



Tabulation:

Input Voltage: $V_{in} = 1V$

TABLE

Parameter	Experimental Value
LOWER CUTOFF FREQUENCY	312.704 Hz
UPPER CUTOFF FREQUENCY	410.661 KHz

RESULT

The lower and upper cutoff frequencies of the CE amplifier were determined successfully.

EXPERIMENT – 7.3

BANDWIDTH ANALYSIS OF COMMON EMITTER AMPLIFIER

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Plot frequency response characteristics.
2. Calculate bandwidth.
3. Analyze amplifier performance.
4. Construct Bode plots.

Aim

To obtain the frequency response characteristics of a CE amplifier and determine its bandwidth.

THEORY

The frequency response of an amplifier describes variation of gain with frequency.

Three frequency regions exist:

Low Frequency Region

Gain decreases due to coupling capacitors.

Mid Frequency Region

Gain remains approximately constant.

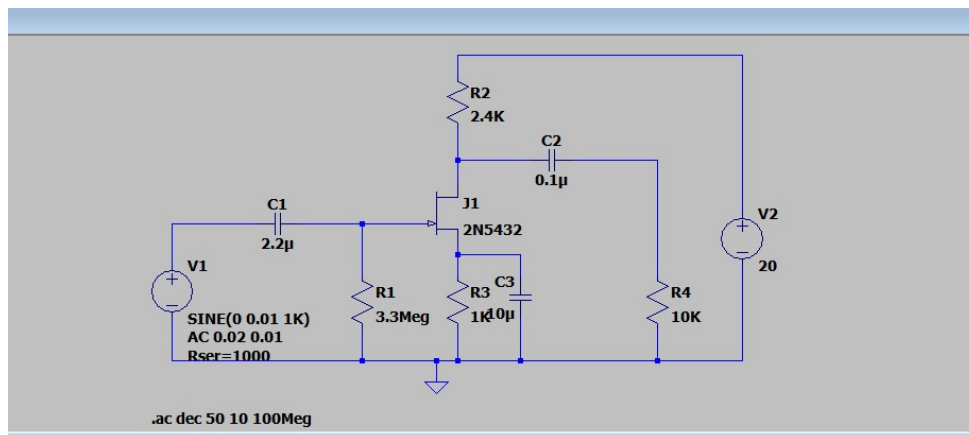
High Frequency Region

Gain decreases because of transistor junction capacitances.

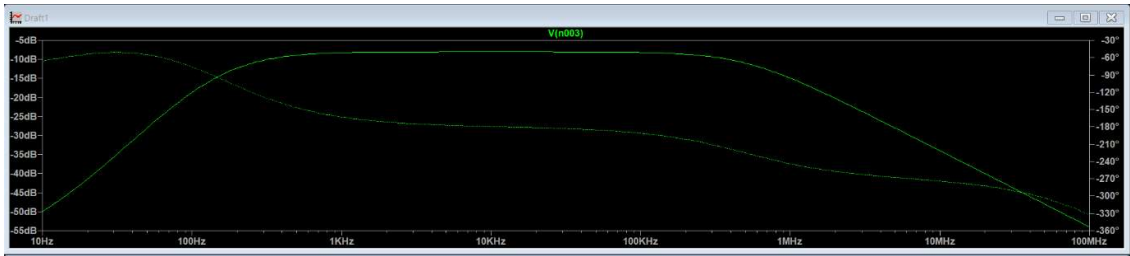
EXPERIMENTAL PROCEDURE

1. Apply constant input voltage.
2. Start at 20 Hz.
3. Record output voltage.
4. Increase frequency gradually.
5. Continue measurements up to 1 MHz.
6. Calculate gain.
7. Convert gain into dB.
8. Plot Bode magnitude response.
9. Determine bandwidth.

Circuit diagram:



Model Graph:



Tabulation:

Input Voltage: $V_{in} = 20 \text{ mV}$

TABLE

Parameter	Experimental Value
LOWER CUTOFF FREQUENCY	288.319 Hz
UPPER CUTOFF FREQUENCY	451.207 KHz
BANDWIDTH	450.918 KHz

RESULT

The frequency response characteristics of the common emitter amplifier were obtained successfully and the bandwidth was determined.